Gauge invariance of a photon

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Gauge Invariance

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- Classical:

$$P^M = (E; P_x, p_y, p_z) \in Classical$$

- Quantum:

$$P^\mu = i\partial^\mu = (i\partial^0; -i\nabla)$$

- For a particle with Charge e:

$$(p^\mu - eA^\mu) = i(\partial^\mu + ieA^\mu)\ldots$$

- Required invariance of the laws of physics under a change of phase.
-

$$A^\mu \Rightarrow A^\mu - \partial^\mu \Lambda$$

- Arbitrary function of the coordinates.

Field Strength Tensor

Field Strength Tensor

- $$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}$$
- $$\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$$
- Maxwell's Equations:
$$\partial_\mu F^{\mu\nu} = -J^\nu$$
- $$\partial_\mu \tilde{F}^{\mu\nu} = 0$$

Lorentz Gauge

Lorentz Gauge and Massless Photons

- Lorentz gauge condition: $\partial_\mu A^\mu = 0$
- $$\partial_\mu(\partial^\mu A^\nu - \partial^\nu A^\mu) = J^\nu$$
- $$\partial_\mu \partial^\mu A^\nu - \partial_\mu \partial^\nu A^\mu = J^\nu$$
- With the Lorentz Gauge, this simplifies to:

$$\partial_\mu \partial^\mu A^\nu = J^\nu$$

- For the massless case ($J^\nu = 0$), this becomes the 4-dimensional wave equation:

$$\Box A^\nu = 0$$

Thank You!!!!!!!!!!!!!

Thank you so much for reading this slide:)))))